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TitleEOS Tricks and Tips That Make Life Easier

Article TypeTech Tips

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Published DateDecember 1, 2021

Updated DateDecember 11, 2025

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Introduction

Arista's Extensible Operating System has several small features that make life in the CLI simpler, quicker, neater, and all-around just easier. Here are a few common practices which Arista's Architects and TAC Engineers use with frequency.

Overview

While a plethora of different tricks and methods exist that make life simpler for network engineers, too many exist to completely cover in this document. If you want a more comprehensive read on the subject, check out Arista Warrior, 2nd Edition by Gary A. Donahue.

Arista's Operational Runbook is a much more extensive guide than this introductory article and a *perfect* start to troubleshoot the most common network problems. Please let your account team know if you cannot locate your copy and they can get you another copy of the Operational Runbook.

Navigational Tricks

Show Active and Show Active All

- Verifies the configuration in the current hierarchy. Use this to check existing configurations without going through a show run.

```
switch(config-router-bgp)#address-family ipv4
switch(config-router-bgp-af)#show active
router bgp 65000
  address-family ipv4
    neighbor border activate
    no neighbor 172.16.13.1 activate
    no neighbor 172.16.13.3 activate
    no neighbor 172.16.13.5 activate
    no neighbor 172.16.13.7 activate
    neighbor 172.16.254.25 activate
    neighbor 172.16.254.27 activate
    neighbor 172.16.254.29 activate
    neighbor 172.16.254.31 activate
    network 172.16.0.103/32
```

- In EOS, adding the **all** command displays both the configured and default configurations.

```
switch(config-router-bgp-af)#show active all
router bgp 65000
  address-family ipv4
    no bgp additional-paths install
    no bgp next-hop address-family ipv6
    no bgp skip rib-install bestpath-selection
    no bgp missing-policy direction in action
    no bgp missing-policy direction out action
    bgp additional-paths receive
    no bgp additional-paths send any
    no bgp next-hop-unchanged
    no bgp route install-map
    no next-hop resolution disabled
    ***snipped***
```

Using “conf” at Any Hierarchical Level

Configuration or simply conf is not just for entering global configuration mode. Use **conf** to return to the global configuration level without entering **exit** multiple times.

```
switch(config-router-bgp)#conf
switch(config)#
```

Acceptance of CIDR Notation

EOS accepts CIDRs, removing the burden of figuring out subnet and wildcard masks.

```
Router ospf 1
Network 192.168.1.0/24 area 0
If you want to put in the extra work, you may but why? Time is a commodity and thus it
switch#(config-if-Et1-3)#router ospf 1
switch#(config-router-ospf)#network 192.168.1.0/24 area 0
switch#(config-router-ospf)#network 192.168.2.0 0.0.0.255 area 0.0.0.0
switch#(config-router-ospf)#show active
router ospf 1
  network 192.168.1.0/24 area 0.0.0.0
  network 192.168.2.0/24 area 0.0.0.0
  max-lsa 12000
switch#(config-router-ospf)#
```

Tab Completion

Using the Tab key will auto-complete custom-built names created such as VRF Instances, ACLs, and BGP Peer Groups. *EVPN_Overlay* is the custom name created below. Tabbing is also done without a carriage return which gives you a cleaner output.

```
switch#(config)#router bgp 65010
switch#(config-router-bgp)#neighbor EVPN_Overlay remote-as 650011
switch#(config-router-bgp)#neighbor EVPN_Overlay ebgp-multihop 3 <- tabbed at the unders
switch#(config)#router bgp 65010
switch#(config-router-bgp)#neighbor EVPN_Overlay <--tabbed at the “_” and no extra lin
switch#(config-router-bgp)#
```

No “do” is needed in config mode

In EOS, the **"do"** parameter is optional for exec-level commands while in configuration mode.

```
Switch(config)#router bgp 65031
Switch(config-router-bgp)#show ip int brief
```

Interface	IP Address	Status	Protocol	MTU
-----	-----	-----	-----	-----
Ethernet49/1	10.139.226.2/31	up	up	8000
Loopback0	10.139.227.228/32	up	up	65535
Management1	172.28.160.181/22	up	up	1500
Port-Channel1	10.139.226.0/31	down	lowerlayerdown	1500
Vlan30	1.1.1.1/24	up	up	1500

Adding “all” to your Show Run

This command will display the running configuration, including any default configurations .
 Caution: do not execute a show run all unless you also use the | **more** so that you can control the scroll rate.

```
show run all | more.
switch#show running all | more
! Command: show running-config all
! device: <removed> (CCS-720XP-24ZY4, EOS-4.27.0F)
!
! boot system flash:/EOS-4.27.0F.swi
!
no enable password
aaa root secret 5 <removed>
aaa authentication policy local allow-nopasssword-remote-login
no username root ssh principal
no aaa authorization policy local default-role
!
username <removed> privilege 15 role network-admin <removed>
username <removed> privilege 15 role network-admin secret 5 <removed>
username <removed> privilege 15 role network-admin secret 5 <removed>
username <removed> privilege 15 role network-admin secret 5 <removed>
!
no hardware access-list ipv4 egress resource sharing routed-interfaces
!
default hardware access-list ipv6 implicit-permit icmpv6
!
--More--
```

Troubleshooting Tips

Watch Diff

Watch Differential - what an amazing tool. It is frustrating and time-consuming to run a show command, run the command a second time, then have to “stare and compare” to see if anything changed. EOS has a feature known as **Watch Diff**. By placing **watch diff** before any show command, the feature will highlight changes in the output.

When debugging, Arista Engineers will often open up a separate CLI session to the same device and deploy **watch diff** for monitoring outputs during testing.

Figure 1. Watch Diff Show Interface Ethernet3 Counters | NZ

```
Every 2.0s: CliShell -s ar -p 15 -c show queue-monitor length statistics
```

Report generated at 2021-10-26 02:05:45

Interface	Traffic Class	Count
Et1	10	1
Et1	1	325
Et5	1	1408
Et7	1	182
Et9	1	1816
Et11	10	2
Et11	1	3251
Et13	10	2
Et13	1	119
Et15	1	746

Figure 2. Watch Diff Show CPU Counters Queue

```
Every 2.0s: CliShell -s ar -p 15 -c show cpu counte... Wed Oct 6 21:03:46 2021 ^
```

Jericho0.0:

CoPP Class	Queue	Pkts	Octets
DropPkts	DropOctets		
Aggregate			

CoppSystemMirroring	Port0	207158	27632695
0	0		

Jericho0.0:
This indicates the sum of per-chip counts of packets permitted and denied by the port-dependent policers.

CoPP Class	Pkts	Octets	Drop
DropPkts	DropOctets		
Aggregate			

Jericho0.1:

CoPP Class	Queue	Pkts	Octets
DropPkts	DropOctets		
Aggregate			

Isolating the most recent syslogs

Depending on how your logging is configured, your logging buffer may contain thousands of entries. EOS helps you display more recent events based upon how far back you wish to display syslogs.

```
switch#show logging last 2 ?  
  days      Show messages in last <N> days  
  hours     Show messages in last <N> hours  
  minutes   Show messages in last <N> minutes  
  seconds   Show messages in last <N> seconds
```

Network Visibility Analyzer or LANZ

Microbursts can result in hard to resolve packet loss. Being able to diagnose the source of microbursts becomes vital in environments where even the smallest packet loss impacts network performance. Arista can not only track the traffic patterns but has been doing this well for many years. Here is a link to understanding LANZ and what it can do for you.

<https://www.arista.com/en/solutions/network-visibility>

(<https://www.arista.com/en/solutions/network-visibility>)

Grep

EOS opens up the Linux beneath to be used in a host of ways that could take years to master. For starters, combining some Linux searches with the CLI is powerful. Here PIM enabled port-channels and interfaces were being searched for.

```
switch#show run all | grep -e 'Port-Channel\|pim\|Ethernet'
default snmp-server enable traps pim
default snmp-server enable traps pim neighbor-loss
interface Ethernet1
    ip dhcp relay information option circuit-id Ethernet1
    no pim ipv4 sparse-mode
    no pim ipv4 bidirectional
    no pim ipv4 border-router
    pim ipv4 hello interval 30
    pim ipv4 hello count 3.5
    pim ipv4 dr-priority 1
    pim ipv4 join-prune interval 60
    pim ipv4 join-prune count 3.5
    no pim ipv4 neighbor filter
    default pim ipv4 bfd
    no pim ipv4 join-prune transport sctp
    no pim ipv4 local-interface
    no pim ipv4 non-dr install-oifs
    no pim ipv6 sparse-mode
    no pim ipv6 border-router
    pim ipv6 hello interval 30
    pim ipv6 hello count 3.5
    pim ipv6 dr-priority 1
    pim ipv6 join-prune interval 60
    pim ipv6 join-prune count 3.5
    no pim ipv6 neighbor filter
    default pim ipv6 bfd
    no pim bsr ipv4 border
```

Linux access and Historical Show Techs

Unless otherwise configured, all Arista devices utilize the underlying Linux to archive a show tech every hour for 100hrs. The newest show tech automatically rewrites over the oldest show tech. Using Linux and not the EOS to run this operation saves both active memory space and CPU cycles. A dedicated partition exists on all Arista flash drives just for this operation.

For devices holding multiple copies of the full internet routing table, you may wish to consider reducing the number of total saved show techs. This is to ensure there is room to store all the data should these files be needed for debugging later. If you have questions on this, you may contact your Arista System Engineer during deployment or Arista TAC.


```
switch#(config)schedule tech-support interval 60 timeout 30 max-log-files 100 command sh
```

Use of Linux to Help Debug

Arista allows full access to the Linux kernel, and all the native features for debugging are still present. Below is a very basic search through the saved show tech files. All this is may be done on a live device with no impact to the data or control planes!

```
switch#bash
Arista Networks EOS shell
[admin@switch ~]$ cd /mnt/flash/schedule/tech-support/
[admin@switch tech-support]$
(zgrep provides the same functionality as grep but does it for zipped files)
[switch tech-support]$
[switch tech-support]$ zgrep -A 2000 -i "show version detail" *.gz | grep -i "aboot-norcal6-6.1.7-135318"
switch_tech-support_2021-10-03.1210.log.gz:Abboot                                Abboot-norcal6-6.1.7-135318
switch_tech-support_2021-10-03.1310.log.gz:Abboot                                Abboot-norcal6-6.1.7-135318
switch_tech-support_2021-10-03.1410.log.gz:Abboot                                Abboot-norcal6-6.1.7-135318
switch_tech-support_2021-10-03.1510.log.gz:Abboot                                Abboot-norcal6-6.1.7-135318
switch_tech-support_2021-10-03.1610.log.gz:Abboot                                Abboot-norcal6-6.1.7-135318
switch_tech-support_2021-10-03.1710.log.gz:Abboot                                Abboot-norcal6-6.1.7-135318
```

Event-monitor

A feature that allows you to monitor changes impacting your network that are seemingly random in nature, making catching the event an automated process. More granular details can be found [here](https://www.arista.com/en/um-eos/eos-event-monitor). (<https://www.arista.com/en/um-eos/eos-event-monitor>)

Related references to check out

[EOS Tips for Power Users](https://arista.my.site.com/AristaCommunity/s/article/introduction-to-managing-eos-devices-annex-b-eos-tips-for-power-users/) (<https://arista.my.site.com/AristaCommunity/s/article/introduction-to-managing-eos-devices-annex-b-eos-tips-for-power-users/>)

[Operational Runbook](#) (Contact your Arista account team if you cannot locate your copy)

[Arista Warrior, 2nd Edition](https://www.oreilly.com/library/view/arista-warrior-2nd/9781491953037/) (<https://www.oreilly.com/library/view/arista-warrior-2nd/9781491953037/>)

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